

INSTALLATION OF WINDOWS 10 VIRTUAL MACHINE

SHREE LR TIWARI ENGINEERING COLLEGE

- REPORT BY MUBASSHERA ZAIDY

1. INTRODUCTION

Windows 10 is still one of the most commonly used operating systems out there, especially in workplaces and enterprise setups. That's why it makes sense to use it as a target system in cybersecurity labs—it reflects the kind of environment attackers and defenders deal with in the real world. Setting it up in a virtual machine gives us a safe space to test things like exploits, malware behavior, and system vulnerabilities without risking the host system. Using a virtual setup also makes it easier to reset, repeat, and tweak configurations as needed. It's flexible, reliable, and works well across different platforms.

2. OBJECTIVE

The main goal of this lab is to set up a functional cybersecurity testing environment by creating a Windows 10 virtual machine using an VHDX file. This virtual machine serves as a target system for practicing ethical hacking techniques such as vulnerability scanning, exploit testing, and malware behavior analysis. The setup provides a hands-on opportunity to explore attack vectors and defensive strategies within a safe, controlled environment.

3. SCOPE

- System requirements and supported platforms :
Covers the basic hardware and software needed to run Windows 10 smoothly across different systems.
- Downloading and installing Windows 10 in a virtual machine :
This includes getting the VHDX file and setting up the VM using a compatible virtualization tool.
- Step-by-step instructions for installation :
Provides clear and simple steps to guide the setup process from start to finish.

4. HARDWARE AND SOFTWARE

- Host Device - Apple MacBook Air (M3 chip)
- Host OS - macOS 16
- Virtualisation - UTM
- Kali Linux(ARM64)
- Windows 10 (VHDX) - Downloaded from Microsoft Insider Preview via Google Drive link.

5. SYSTEM REQUIREMENTS

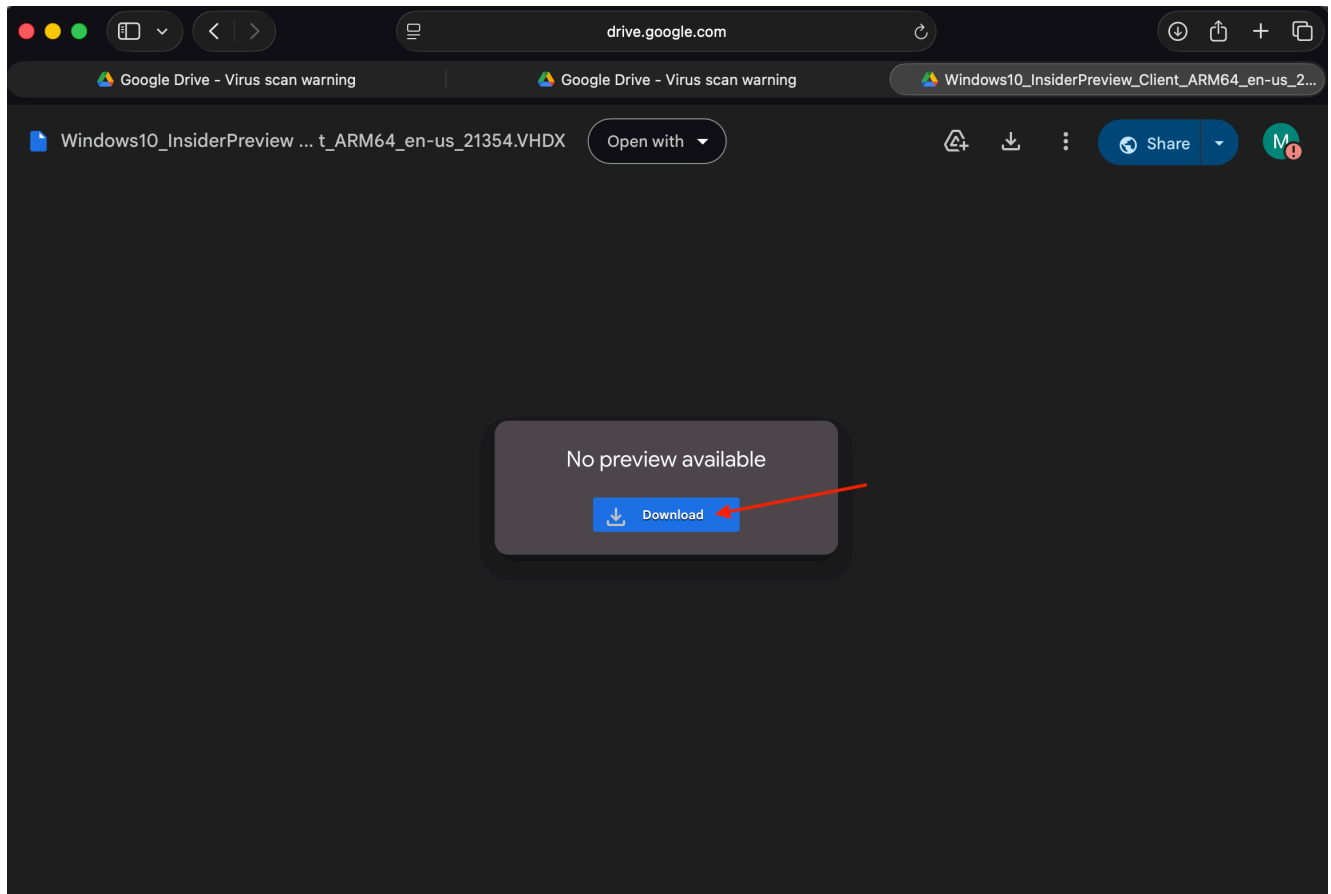
REQUIREMENTS	MINIMUM SPECIFICATION
Processor	1 GHz or faster, 64 bit
RAM	2 GB (4 GB recommended)
Storage	25 GB minimum

6. SUPPORTED OPERATING SYSTEMS

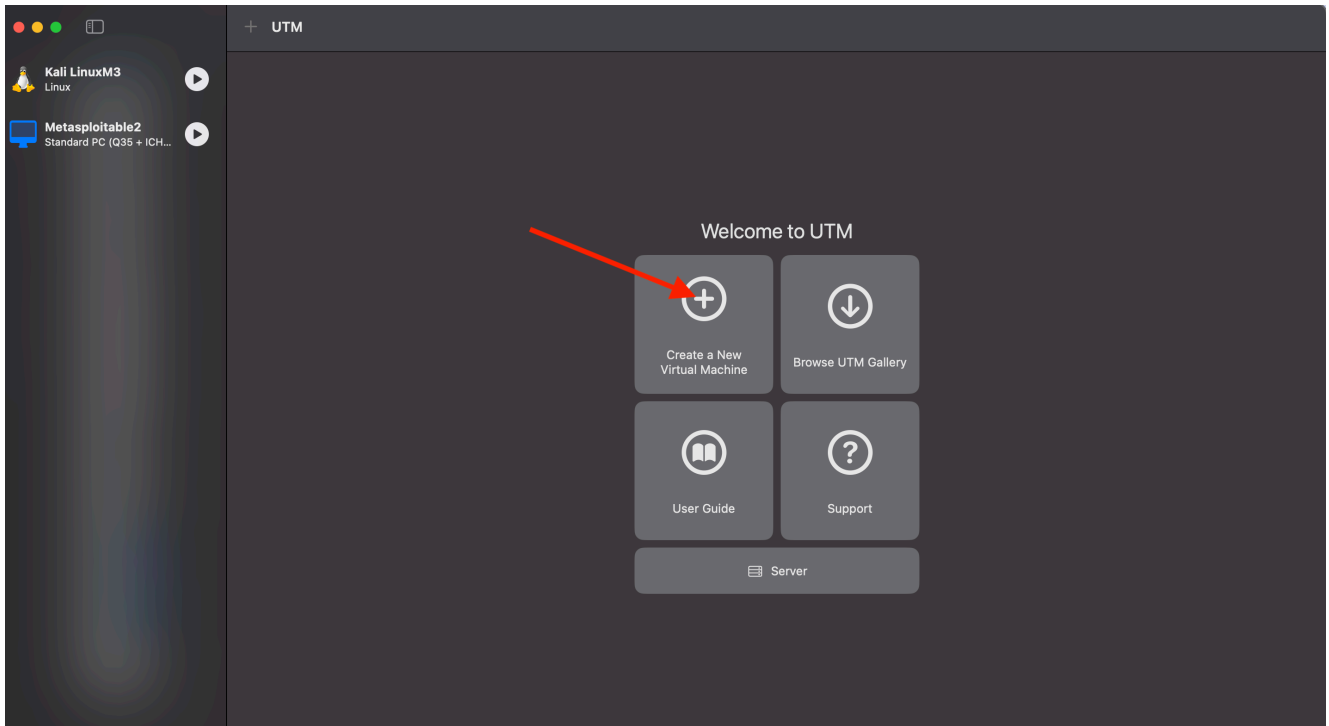
- Windows 10/11
- macOS (Intel or Apple Silicon: M1, M2, M3)
- Linux (Ubuntu, Fedora, Debian, etc.)
- ChromeOS
- Cloud platforms (AWS, Azure, GCP)

7. INSTALLATION STEPS FOR WINDOWS VM

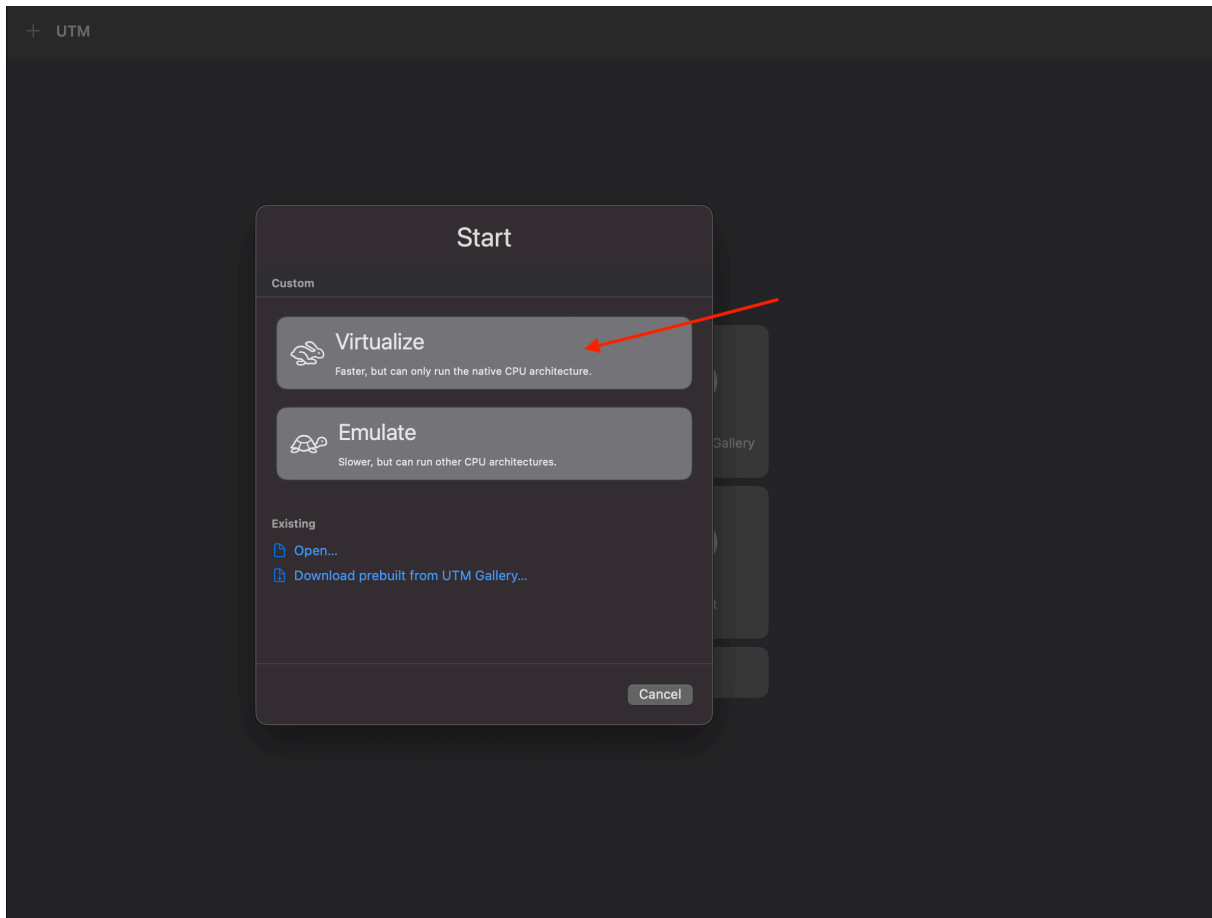
STEP 1 : Download the Windows 10 VHDX file from https://drive.google.com/file/d/1kY3Xlt9TQYtjODUR2p1R_2G_GePB0h5K/view



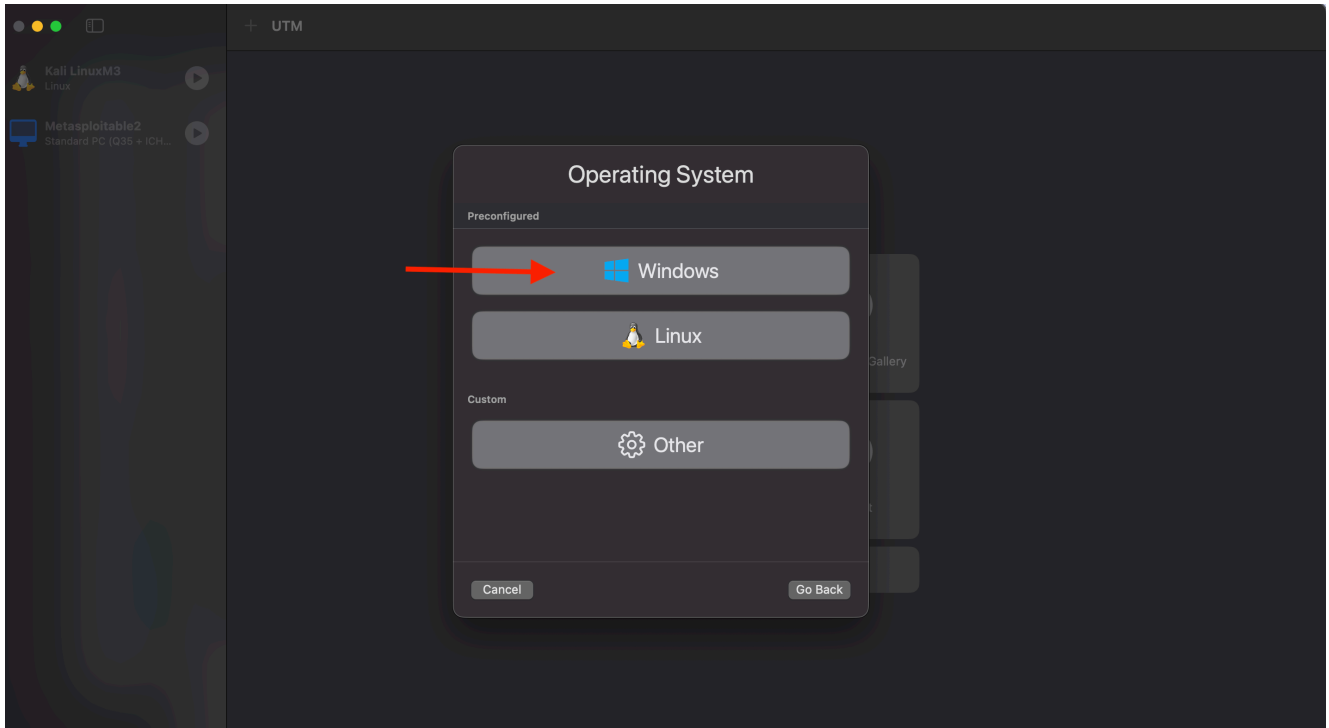
STEP 2 : Once downloaded, open UTM and click on “Create a New Virtual machine”



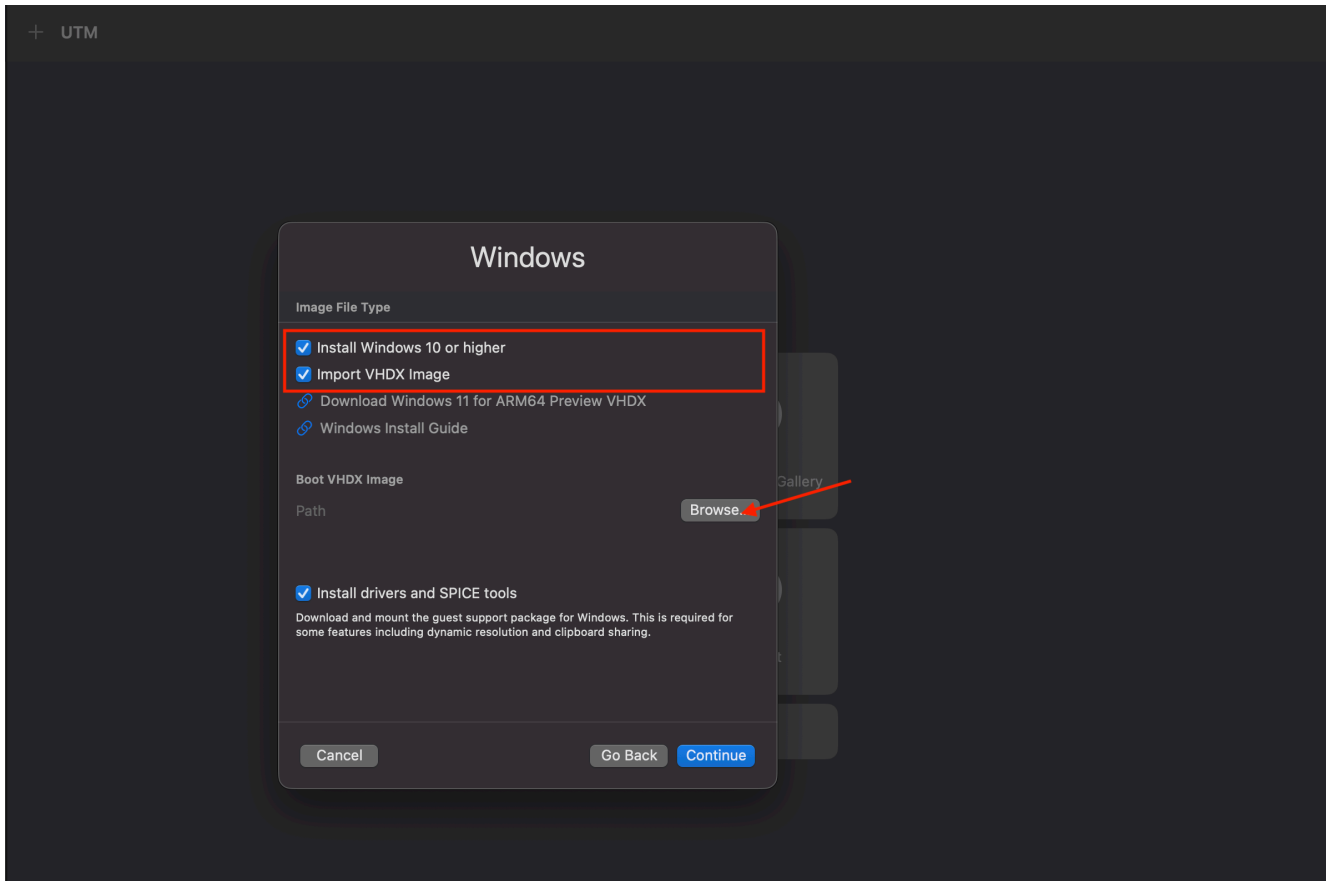
STEP 3 : Click on “Virtualize”.



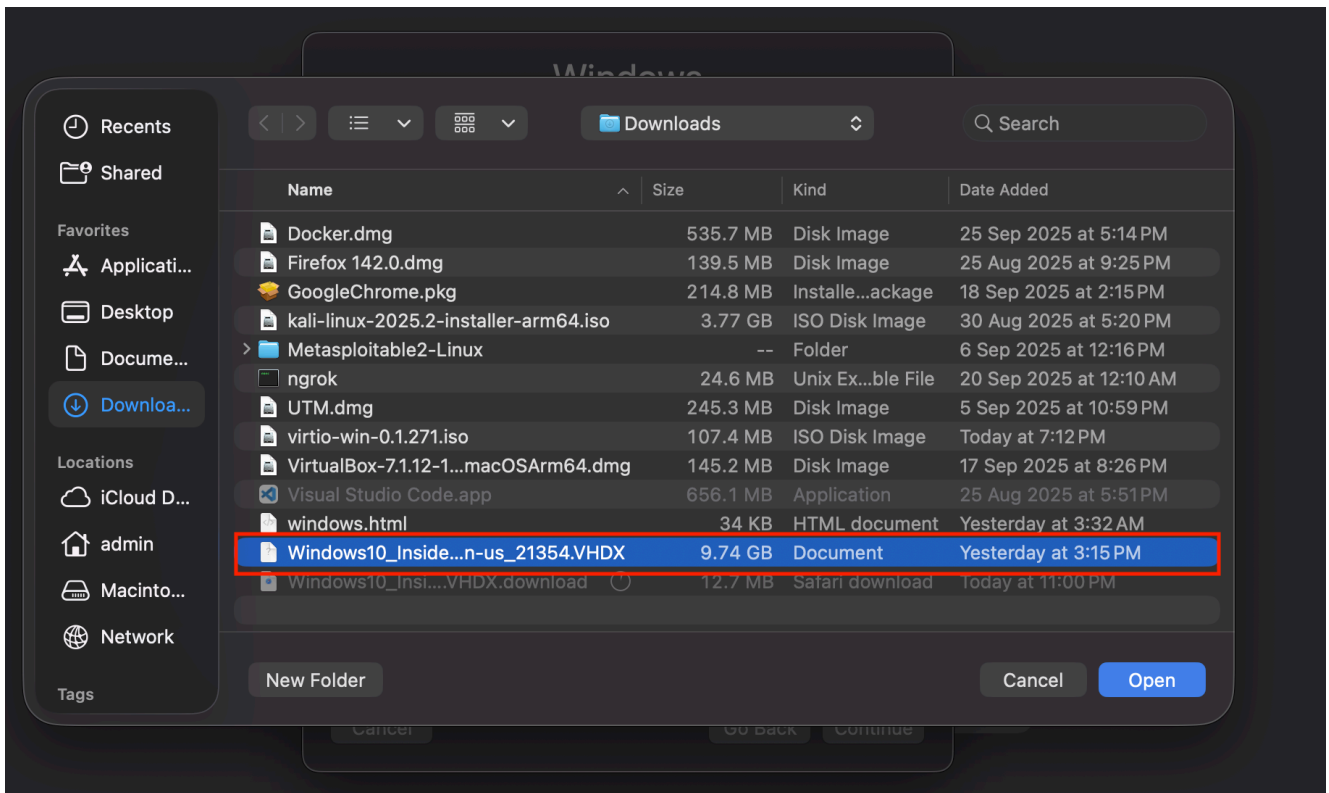
STEP 4 : Click on “Windows”.



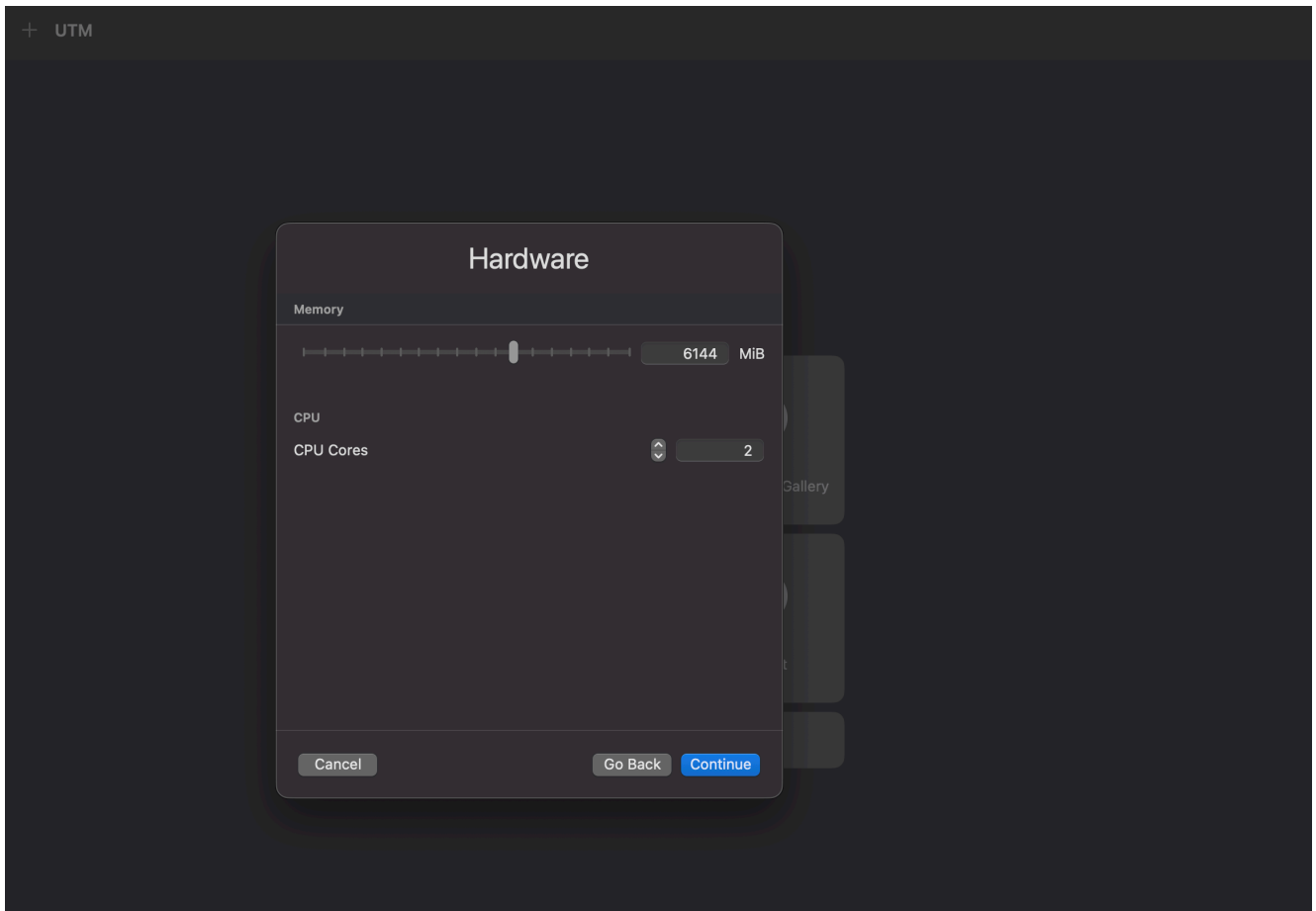
STEP 5 : Enable “Install Windows 10 and Higher” and “Import VHDX image”, then click on “Browse” to select the boot image.



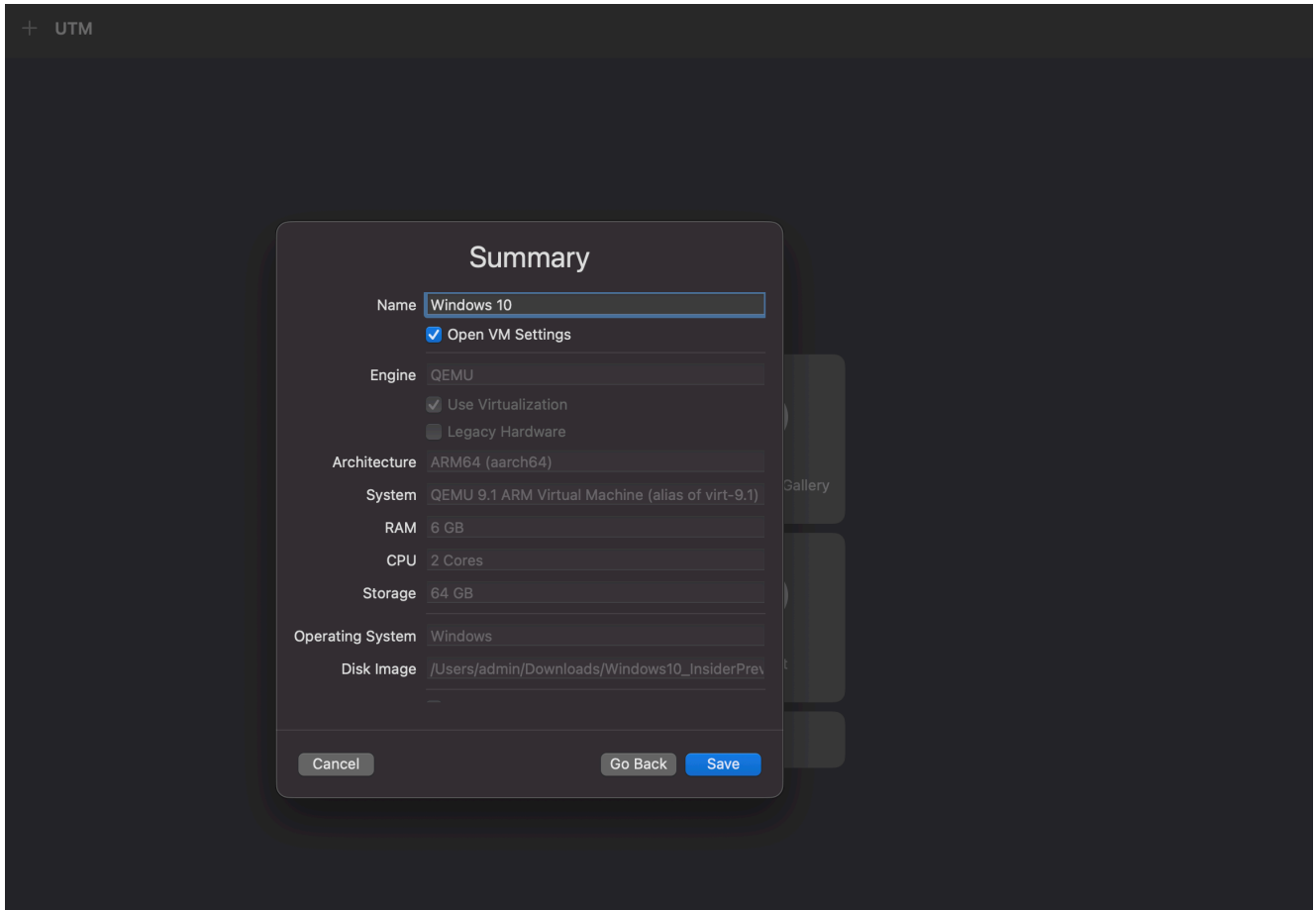
STEP 6 : Locate and select the Windows 10 VHDX file, then click on “Open” to attach it as the virtual disk.



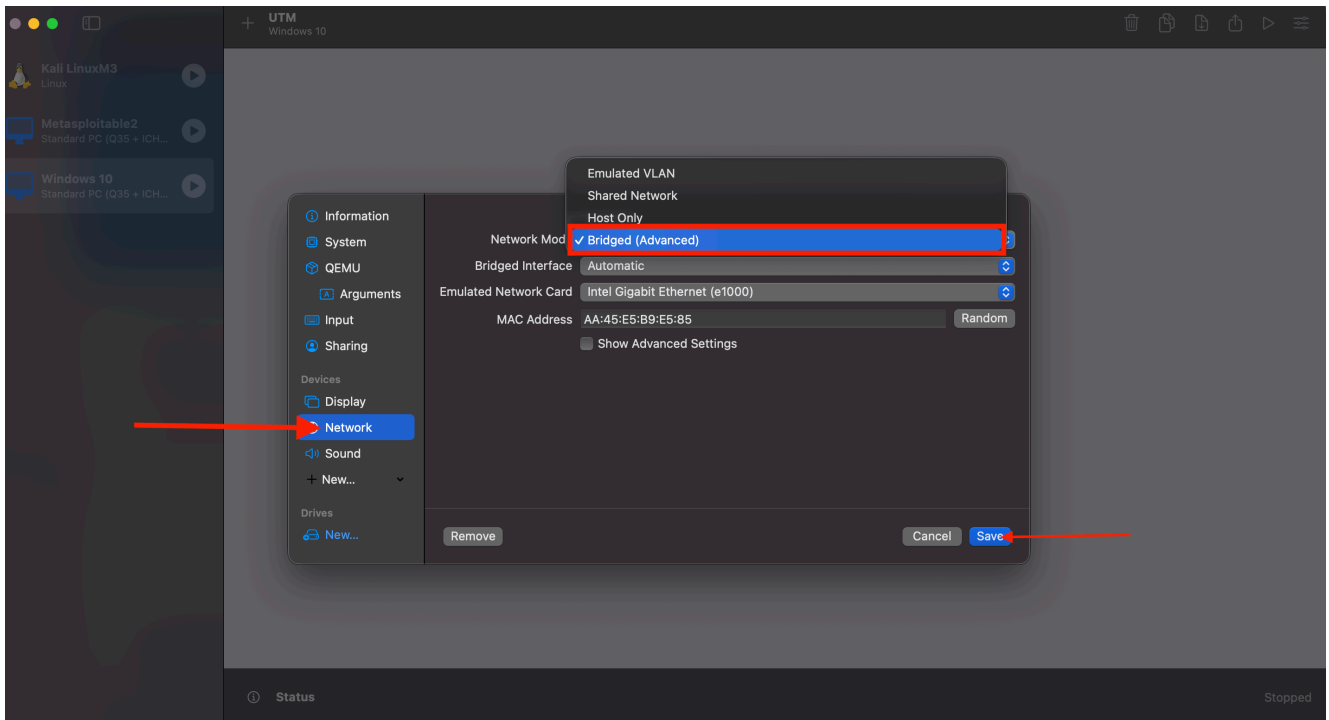
STEP 7 : In Hardware configuration allocate memory and CPU cores in accordance with your host system's specifications , and click on "Continue"



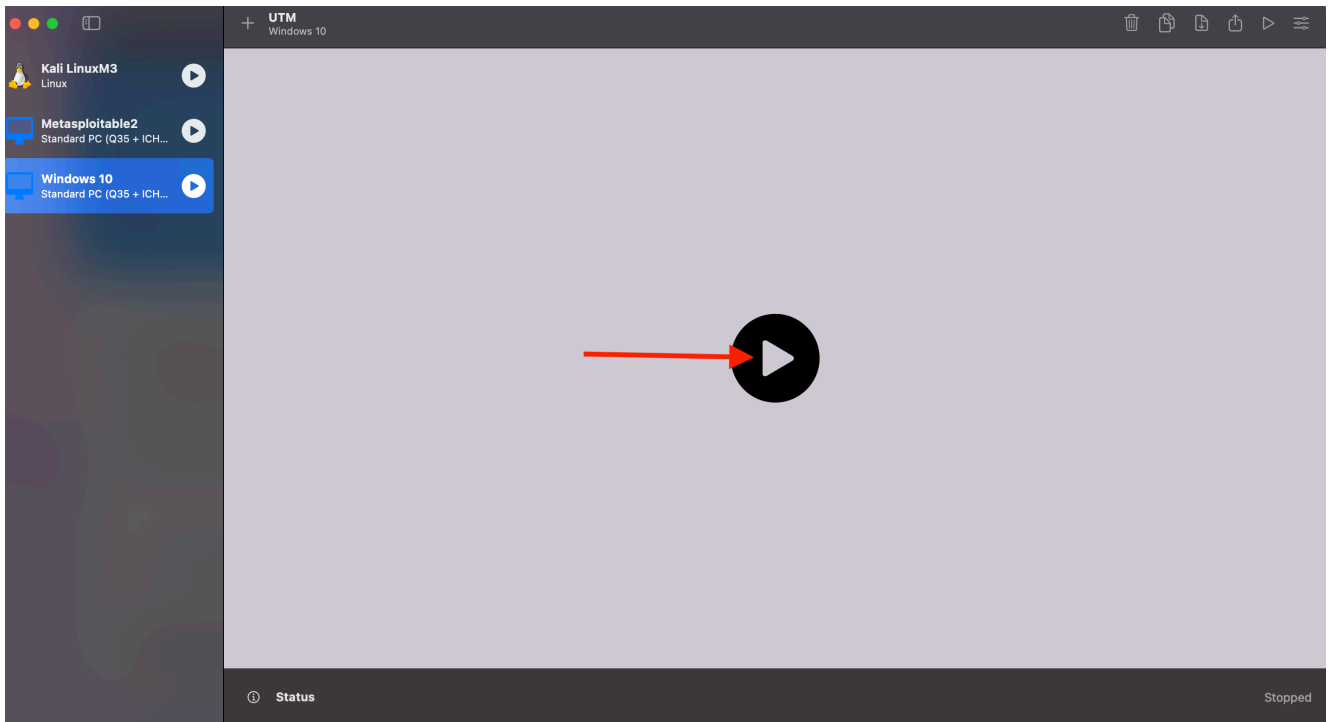
STEP 8 : Assign a name to your virtual machine, enable the “Open VM Settings” option, and click on “Save”



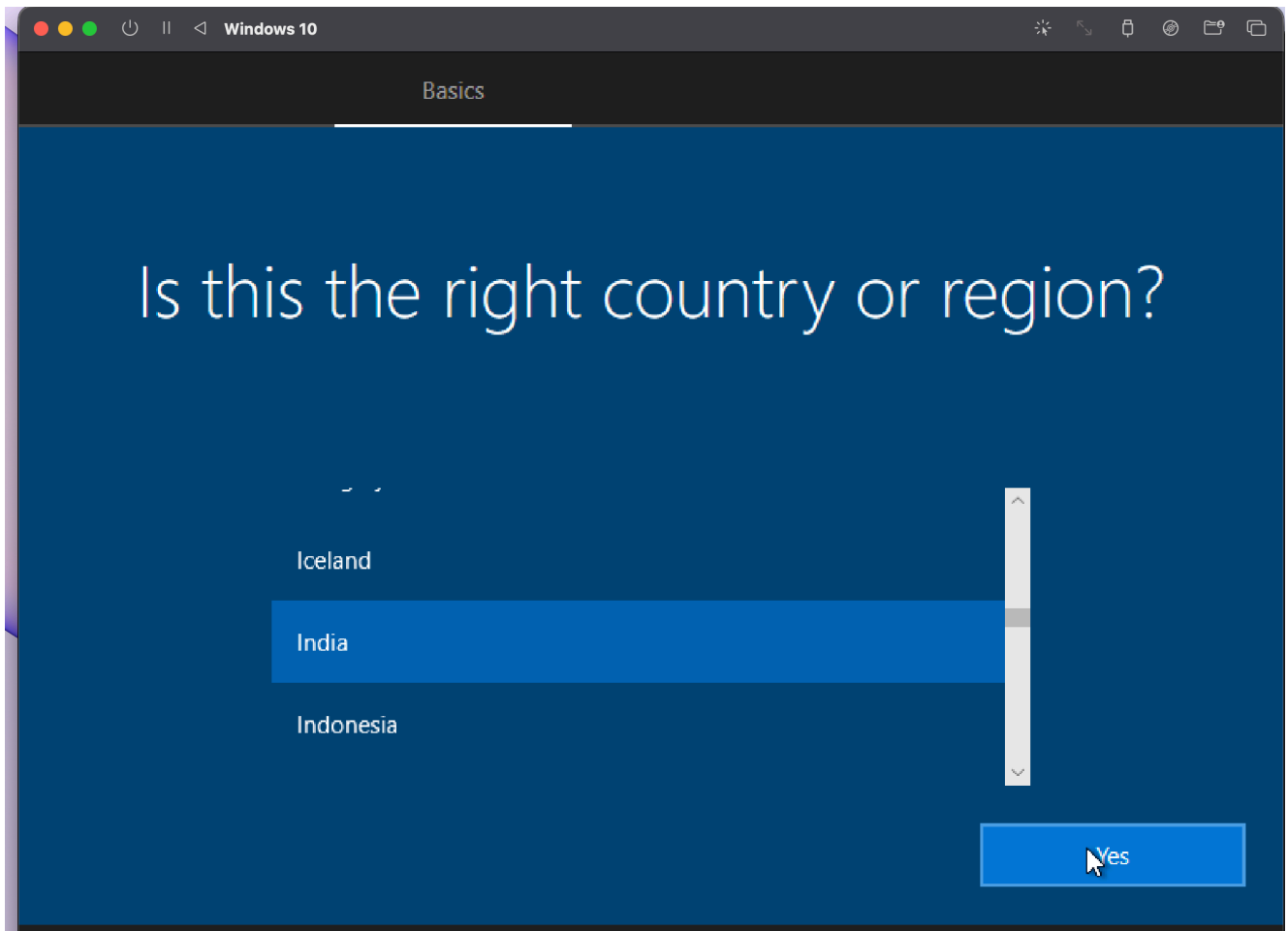
STEP 9 : In the Settings menu, navigate to Network, then select “Bridged” under the Network Mode and click on “Save”.



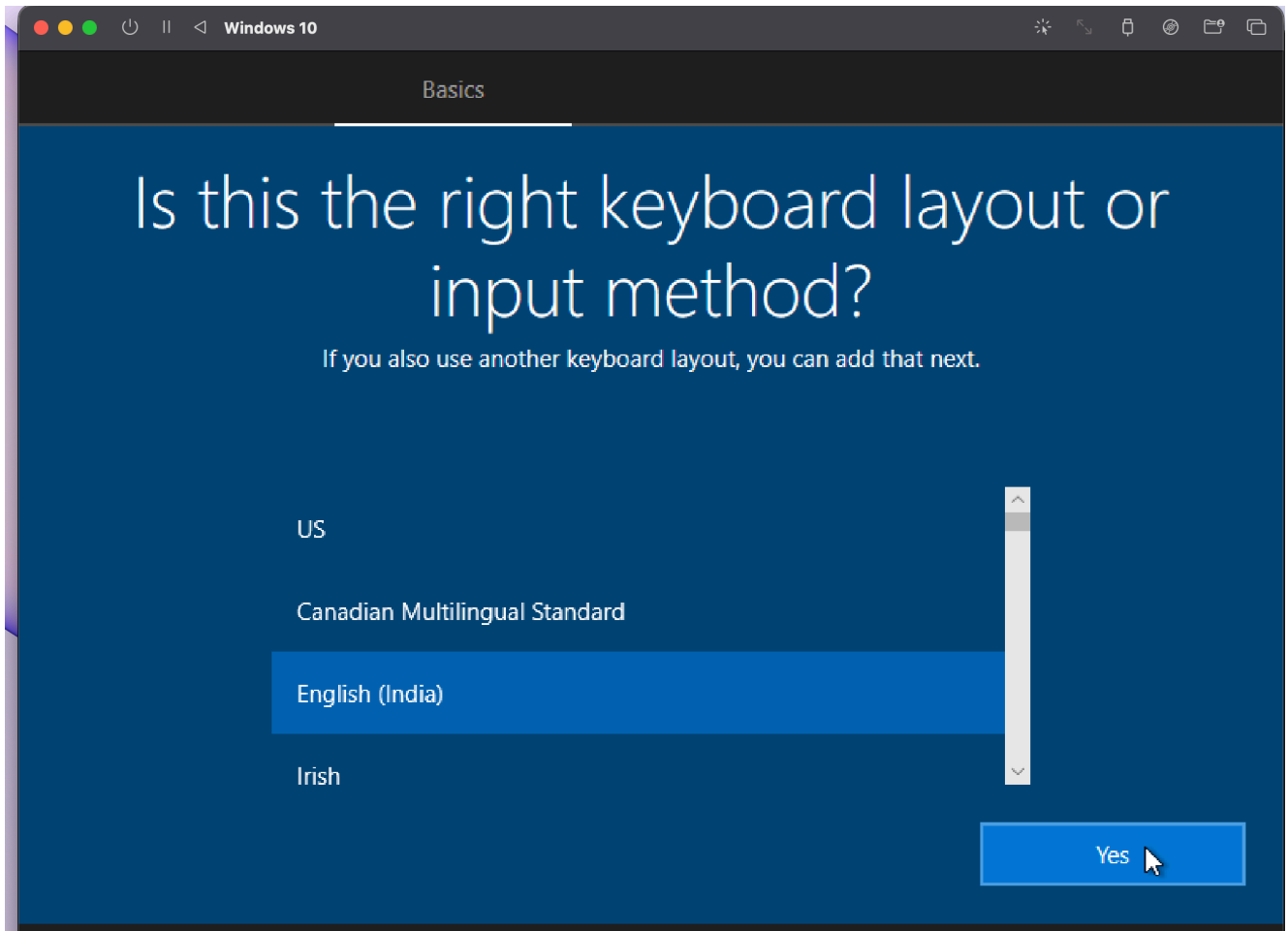
STEP 10 : Click on play button to start the Windows 10 VM



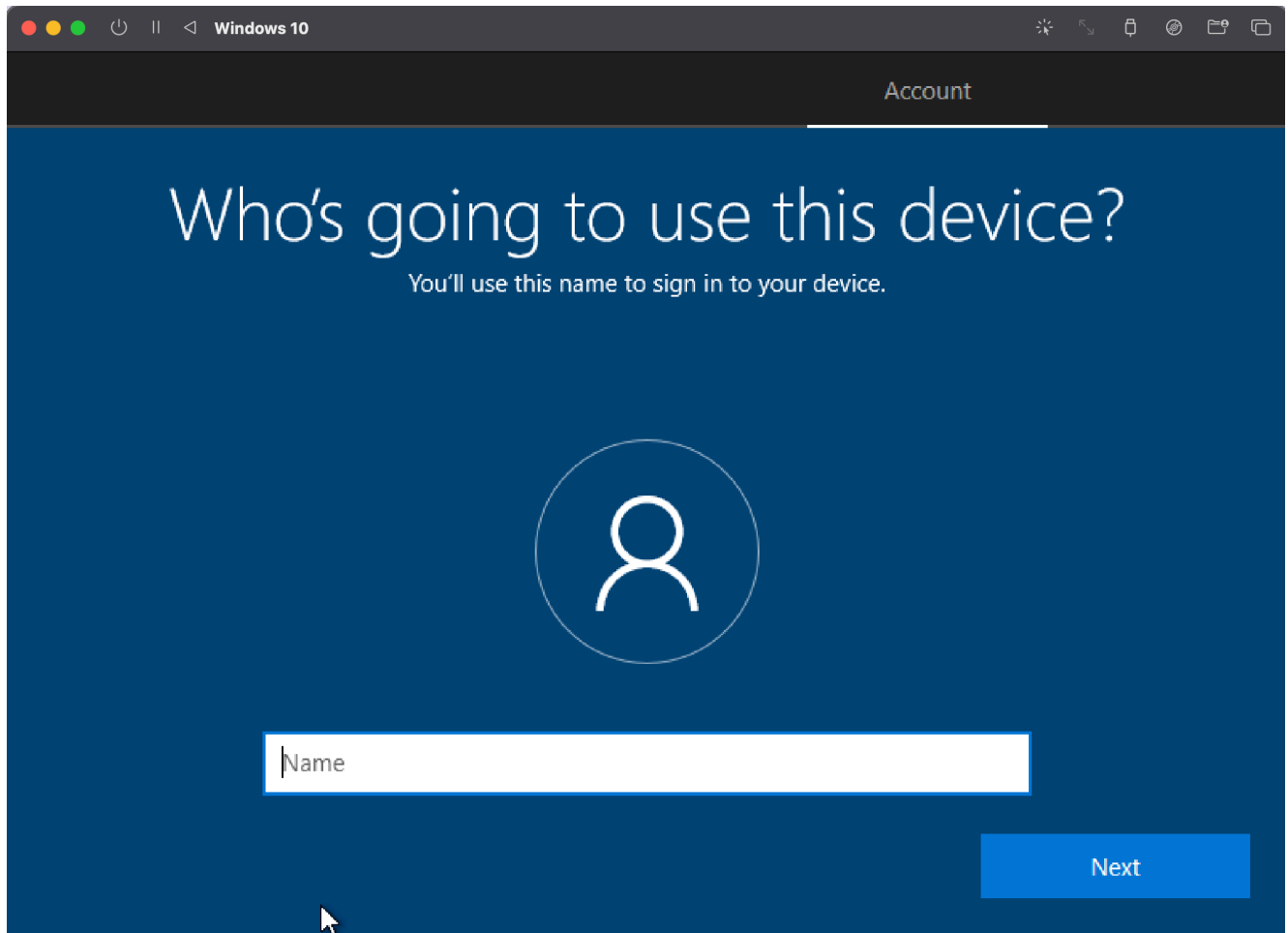
STEP 11 : Once started , select your country or region and click on “Yes”



STEP 12 : Select the right keyboard layout and click on “Yes”



STEP 13 : Type your name to sign in to your device and click on “Next”



The image shows a Windows 10 account setup window. The title bar at the top says "Windows 10" and includes standard window controls. Below the title bar, the word "Account" is centered. The main background is dark blue. The text "Who's going to use this device?" is prominently displayed in white, with a subtitle "You'll use this name to sign in to your device." below it. In the center is a white circular icon representing a person. Below this is a white rectangular text input field with the placeholder text "Name". At the bottom right is a blue button labeled "Next". A mouse cursor is visible near the bottom left of the window.

Windows 10

Account

Who's going to use this device?

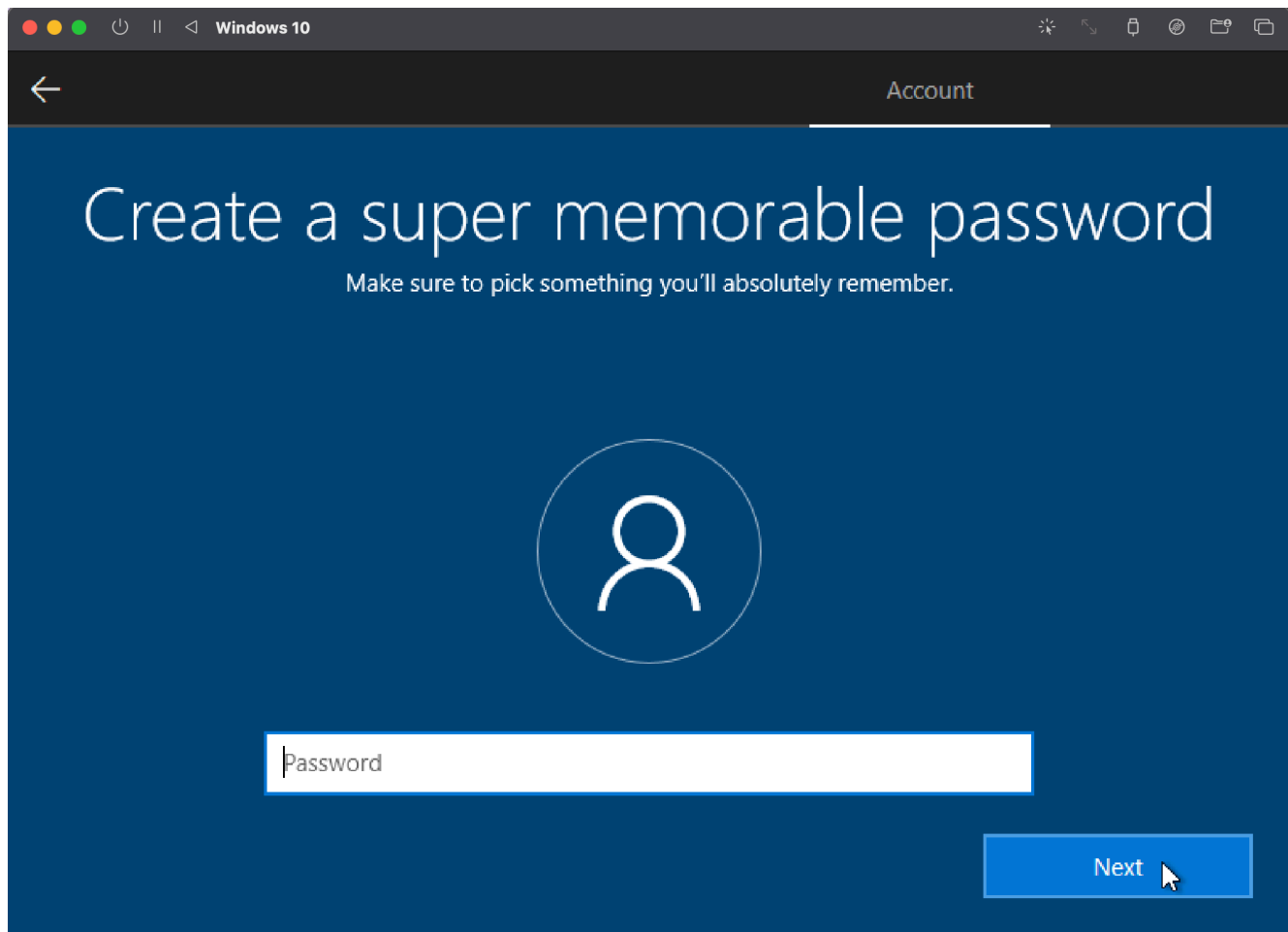
You'll use this name to sign in to your device.



Name

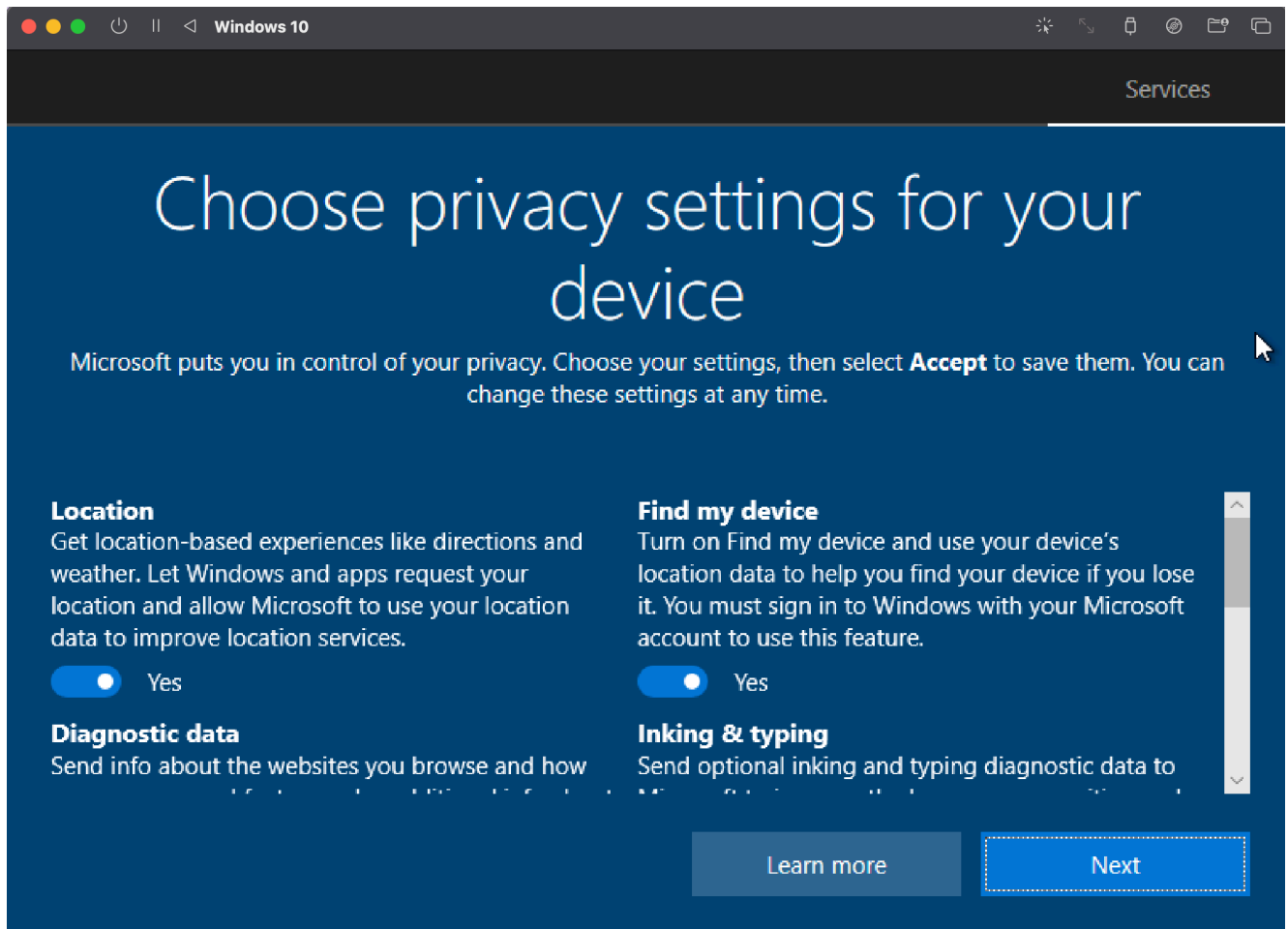
Next

STEP 14 : Create your password to sign in to your device and click on “Next”.

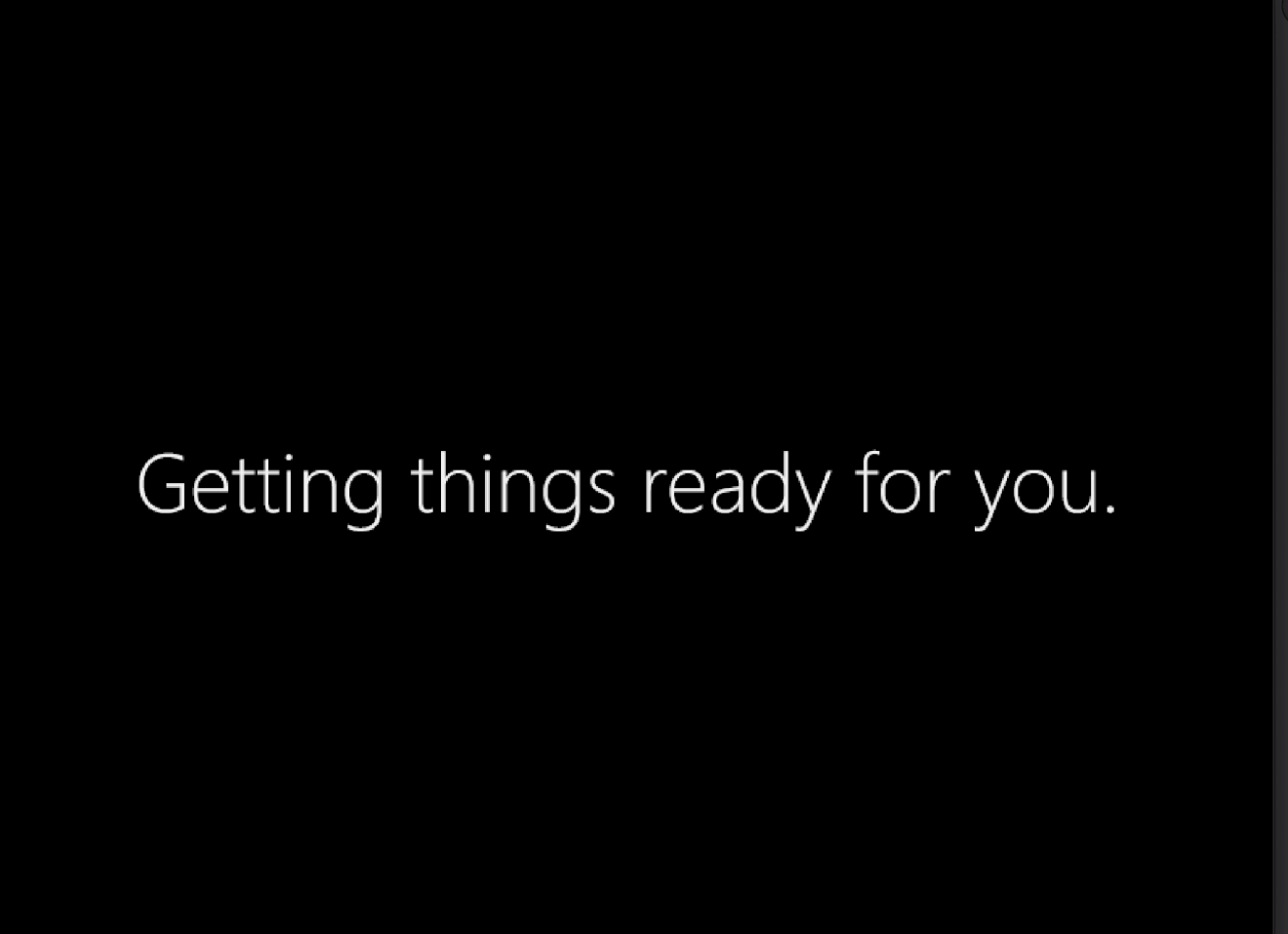


The screenshot shows the Windows 10 account creation interface. At the top, there is a dark header bar with a back arrow on the left and the word "Account" on the right. Below the header, the main background is a solid blue color. The text "Create a super memorable password" is displayed in large white font, followed by the instruction "Make sure to pick something you'll absolutely remember." in a smaller white font. In the center, there is a white circular icon representing a person. Below this icon is a white rectangular input field with the placeholder text "Password". At the bottom right, there is a blue button with the word "Next" in white text, and a mouse cursor is pointing at it.

STEP 15 : Go through the privacy settings and click on “Next”

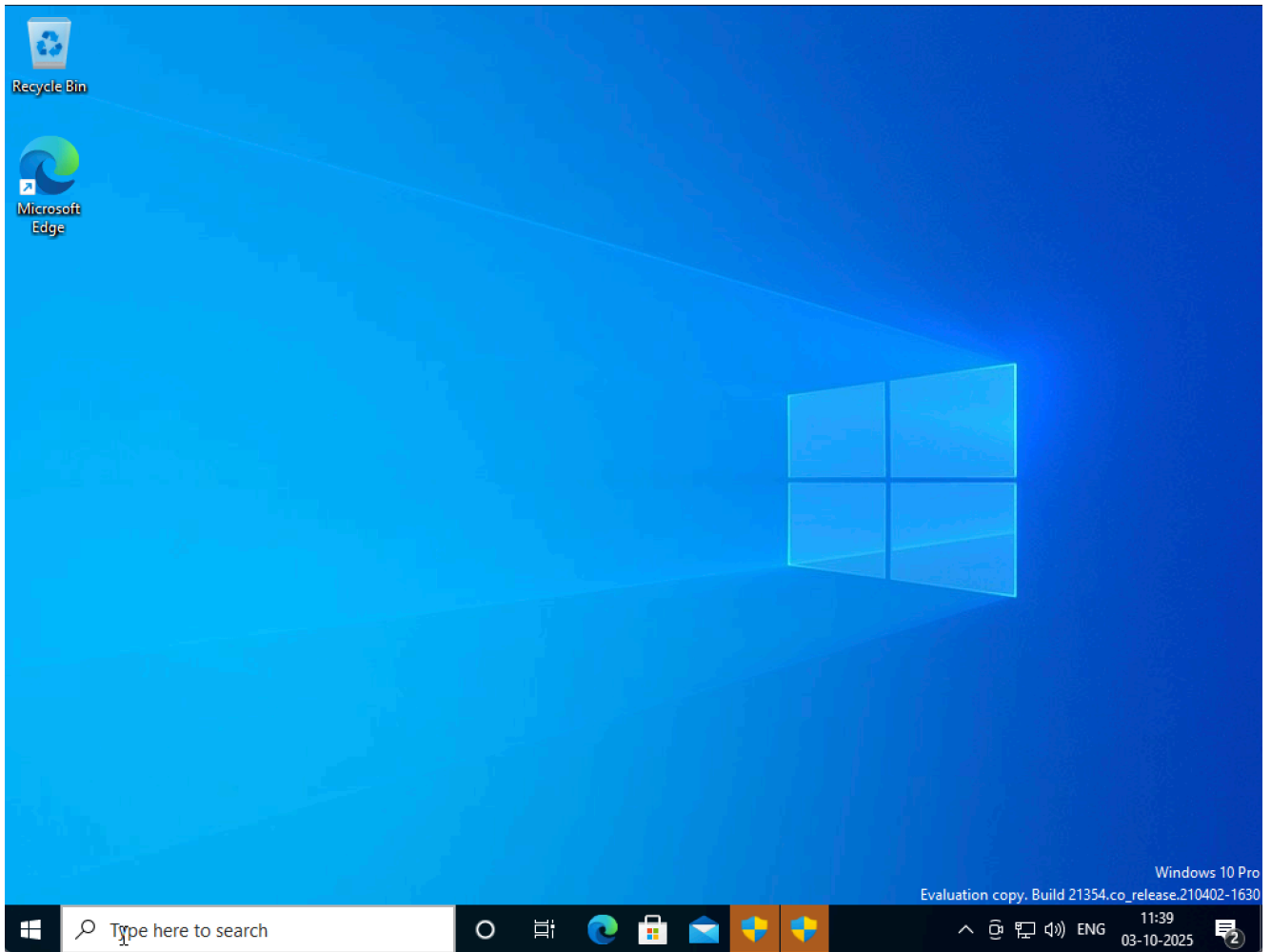


STEP 16 : Wait for Windows to initialize the environment.



Getting things ready for you.

STEP 17 : Once initialization is completed, the Windows desktop environment will be displayed.



TROUBLESHOOTING TIPS

- **VM doesn't boot:** Ensure the VHDX file is ARM64-compatible and properly attached.
- **No network access:** 1) Double-check that "Bridged" mode is selected and your host network is active.
2) Install SPICE tools via UTM's guest support package.
- **Performance lag:** Reduce RAM or CPU allocation if your host system is under strain.
- **Clipboard sharing not working:** Install SPICE tools via UTM's guest support package.

CONCLUSION

By following all the steps in this report, users can easily set up a Windows 10 virtual machine on a Mac using UTM. The process is straightforward and works well with Apple Silicon, making it accessible for students and professionals alike. Once installed, the VM provides a safe and flexible environment for cybersecurity testing, allowing users to explore ethical hacking techniques without affecting their main system. This setup is reliable, repeatable, and well-suited for hands-on learning in cybersecurity labs.